

Reflection and Transmission of EM wave at interfaces

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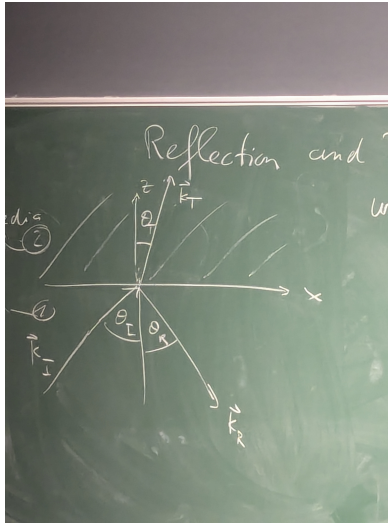


Figure 1:

We want to use the integral versions of Maxwell's equations

$$\int_S \vec{D} \cdot d\vec{S} = Q_{free, enclosed} = 0 \quad (1)$$

Vanishes only if dielectric. S is close surface that encloses the volume V.

$$\int_{\partial S} \vec{E} \cdot d\vec{r} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{S} \quad (2)$$

where ∂S is the boundary of an open surface.

$$\int_{\partial S} \vec{H} \cdot d\vec{r} = I_{encl, free} + \frac{d}{dt} \int_S \vec{D} \cdot d\vec{S} \quad (3)$$

where $I_{enc} = 0$ for a dielectric.

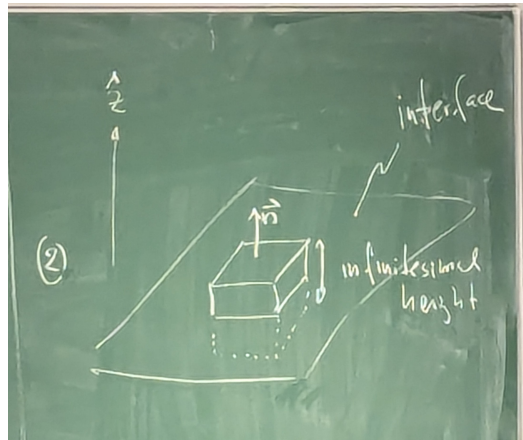


Figure 2:

Here the closed surface is the surface of the box. We are only concerned with up and down surfaces as the box has infinitesimal height. It faces 2 different media above and below.

$$\vec{n} \cdot (\vec{D}_2 - \vec{D}_1) = \sigma_{free} a = 0 \quad (4)$$

$$(5)$$

$$\vec{D}_1^\perp = \vec{D}_2^\perp \quad (6)$$

Analogous for $\vec{B}$

$$\vec{B}_1^\perp = \vec{B}_2^\perp \quad (7)$$

Similarly (Griffiths)

$$\vec{H}_1^\parallel = \vec{H}_2^\parallel \leftrightarrow \frac{1}{\mu_1} \vec{B}_1^\parallel = \frac{1}{\mu_2} \vec{B}_2^\parallel \quad (8)$$

ONLY HOLDS in linear media.
in our geometry $\parallel \hat{x}$, $\hat{y}$ and $\perp \hat{z}$

Summary

$$\epsilon_1 \vec{E}_1^\perp = \epsilon_2 \vec{E}_2^\perp \quad (9)$$

$$\vec{B}_1^\perp = \vec{B}_2^\perp \quad (10)$$

$$\vec{E}_1^\parallel = \vec{E}_2^\parallel \quad (11)$$

$$\frac{1}{\mu_1} \vec{B}_1^\parallel = \frac{1}{\mu_2} \vec{B}_2^\parallel \quad (12)$$

Consequences

$$\vec{E}_{0,I}^\parallel e^{i(\vec{k}\vec{x} - \omega_I t)} \Big|_{z=0} + \vec{E}_{0,R}^\parallel e^{i(\vec{k}\vec{x} - \omega_R t)} \Big|_{z=0} = \vec{E}_{0,T}^\parallel e^{i(\vec{k}\vec{x} - \omega_T t)} \Big|_{z=0} \quad (13)$$

and

$$\epsilon_1 \left(\vec{E}_{0,I}^\perp e^{i(\vec{k}\vec{x} - \omega_I t)} \Big|_{z=0} + \vec{E}_{0,R}^\perp e^{i(\vec{k}\vec{x} - \omega_R t)} \Big|_{z=0} \right) = \epsilon_2 \vec{E}_{0,T}^\perp e^{i(\vec{k}\vec{x} - \omega_T t)} \Big|_{z=0} \quad (14)$$

These must hold for any t and any $\vec{x} = (x, y, 0)^T$.

Not $\vec{x} = 0 \implies \omega_I = \omega_R = \omega_T = \omega$

$\rightarrow$ no change in frequency

If we set $t = 0$

$$\vec{k}_I \vec{x} \Big|_{z=0} = \vec{k}_R \vec{x} \Big|_{z=0} = \vec{k}_T \vec{x} \Big|_{z=0} \quad (15)$$

Choose incident wave vector in the $x - z$ plane

$$\vec{K}_I = k_I (\sin \theta_I \hat{x} + \cos \theta_I \hat{z}) \quad (16)$$

$$\vec{k}_R = k_R (\sin \theta_R \cos \phi_R \hat{x} + \sin \theta_R \sin \phi_R \hat{y} - \cos \theta_R \hat{z}) \quad (17)$$

$$\vec{k}_T = k_T (\sin \theta_R \cos \phi_R \hat{x} + \sin \theta_R \sin \phi_R \hat{y} + \cos \theta_R \hat{z}) \quad (18)$$

set

$$\vec{x} \Big|_{z=0} = \hat{x} \quad (19)$$

$$\vec{x} \cdot \vec{k}_I = k_I \sin \theta_I = k_R \sin \theta_R \cos \phi_R = k_T \sin \theta_T \cos \phi_T \quad (20)$$

set

$$\vec{x}\Big|_{z=0} = \hat{y} \quad (21)$$

$$0 = k_R \sin \theta_R \sin \phi_R = k_T \sin \theta_T \sin \phi_T \quad (22)$$

so

$$\phi_R = \phi_T = 0 \implies \vec{k}_I, \vec{k}_R, \vec{k}_T \quad (23)$$

are in $x - z$ plane. (Plane of incidence)
1st law of geometric optics

$$k_R \sin \theta_R = k_T \sin \theta_T = k_I \sin \theta_I \quad (24)$$